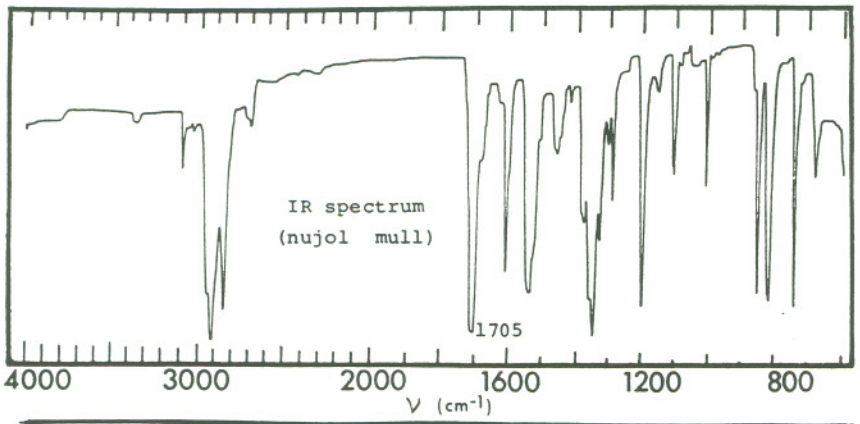
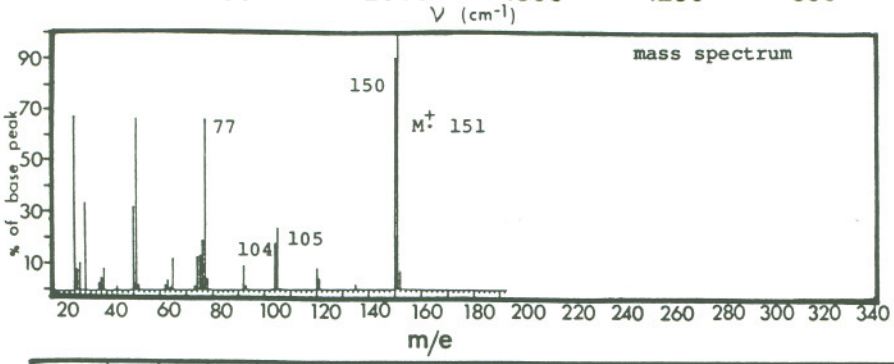
Problem 1

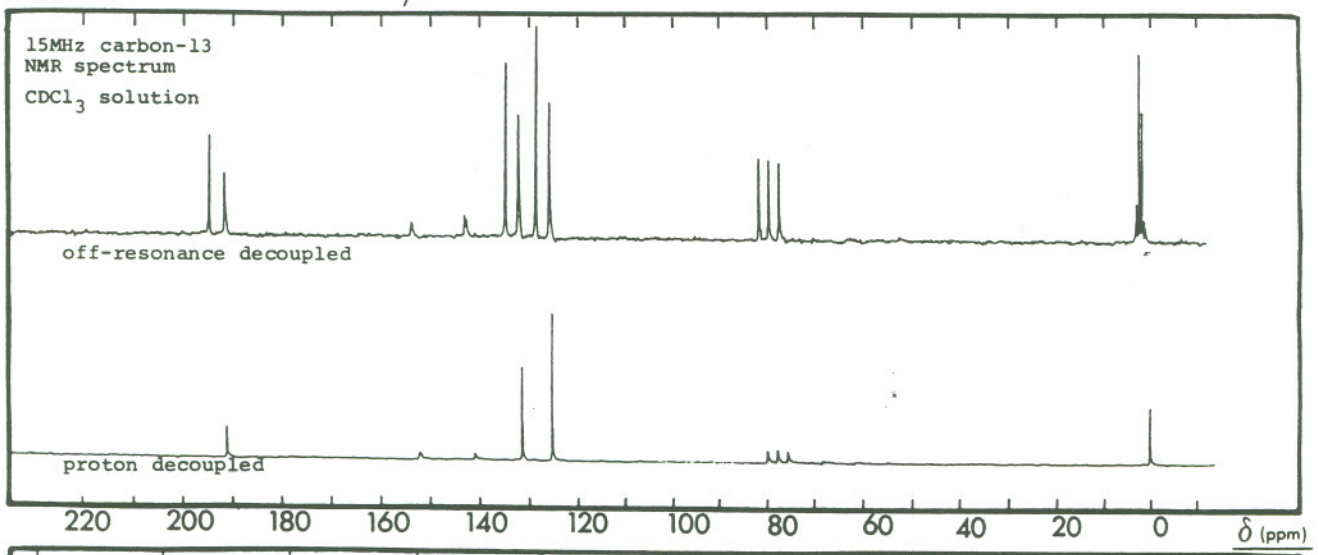
ORGANIC CUMULATIVE EXAM

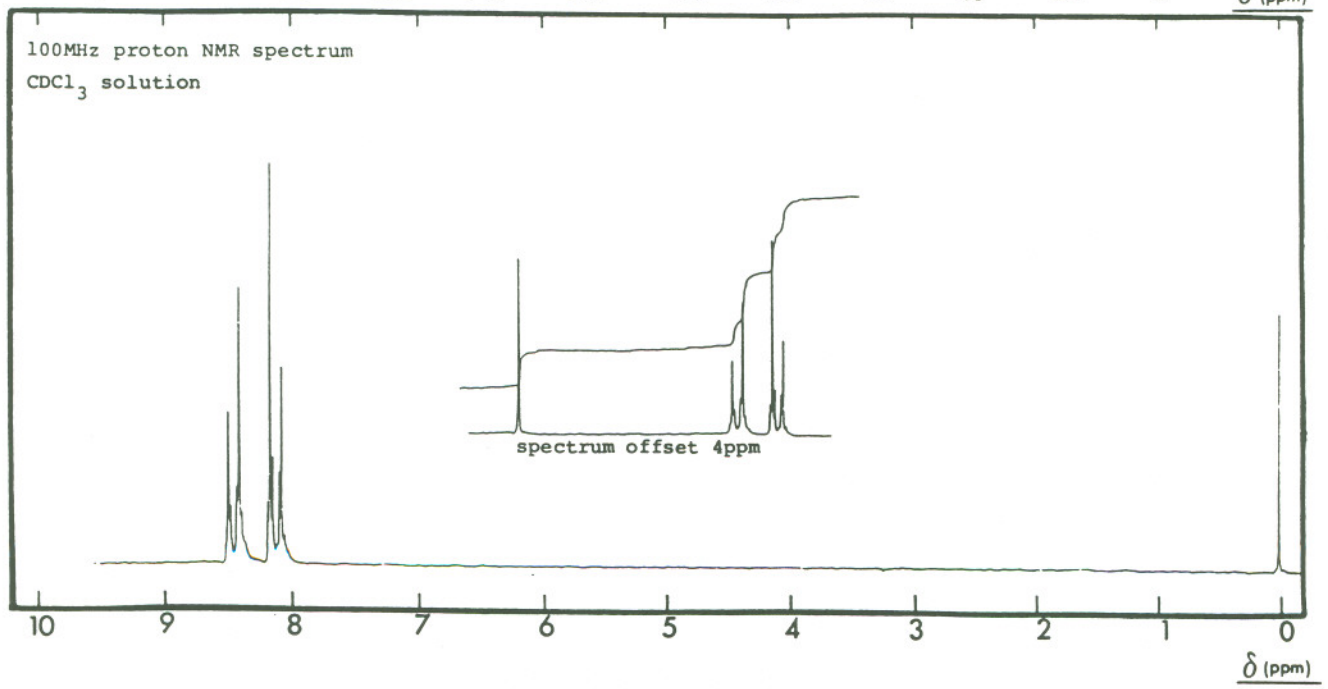
December 3, 1994

Give a structure consistent with each of the following sets of spectra.

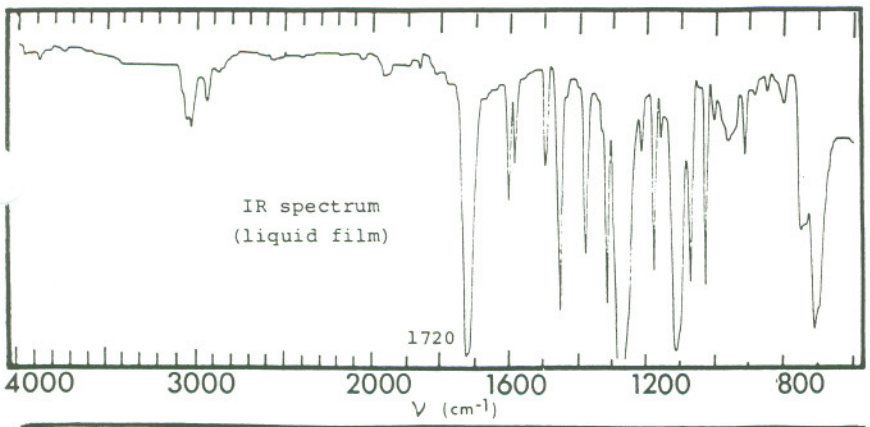


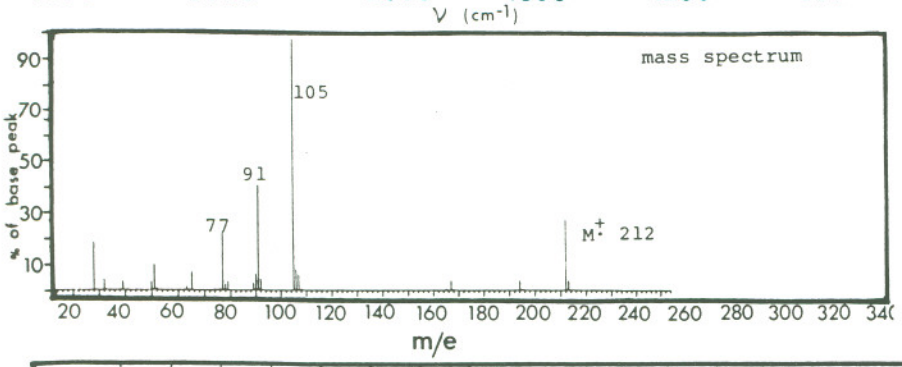


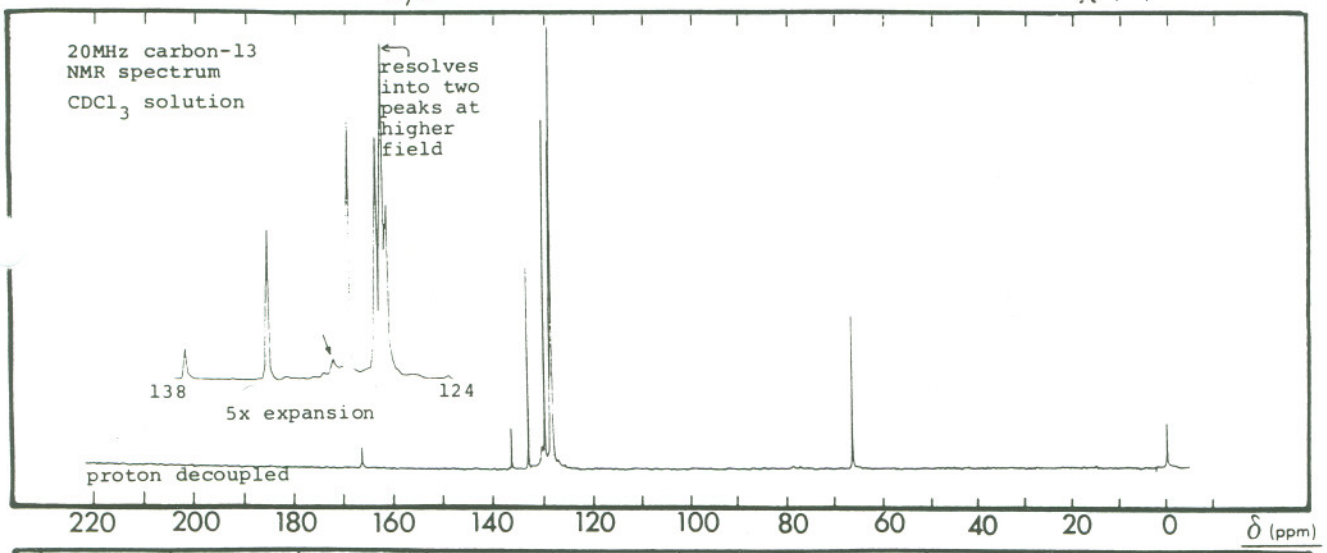


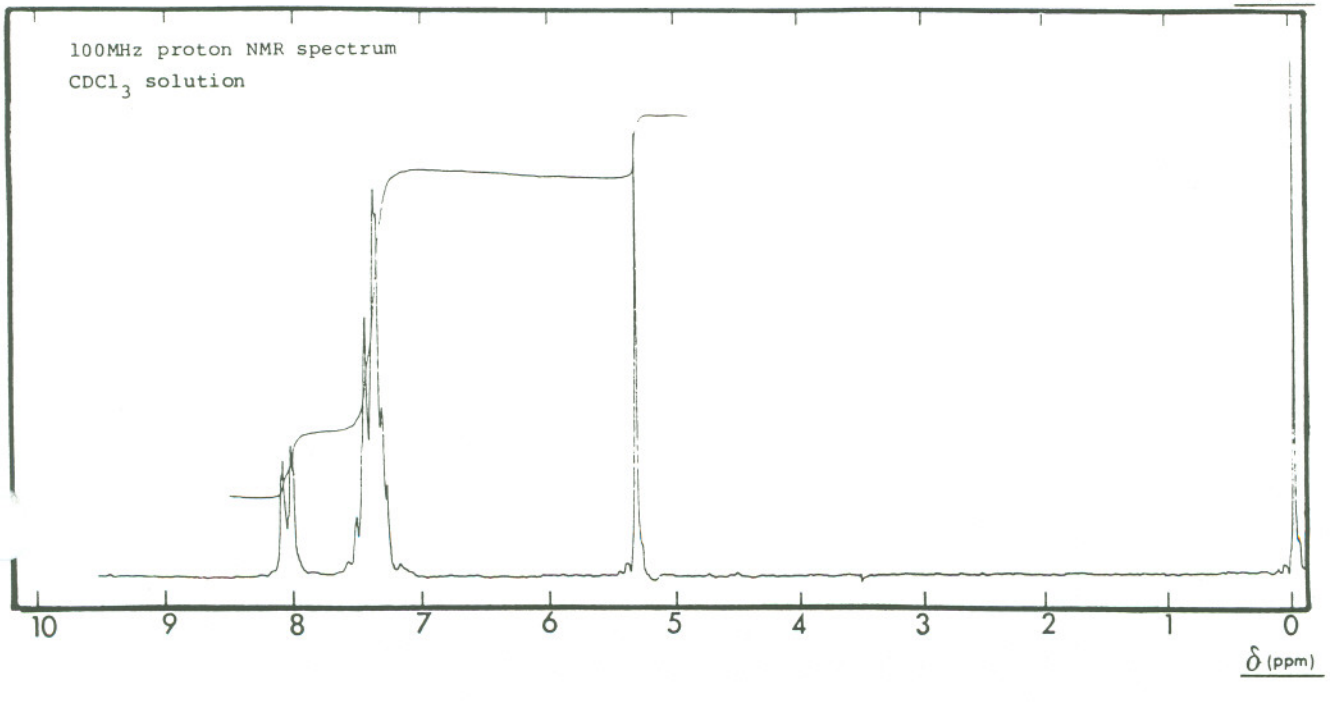


Problem 2

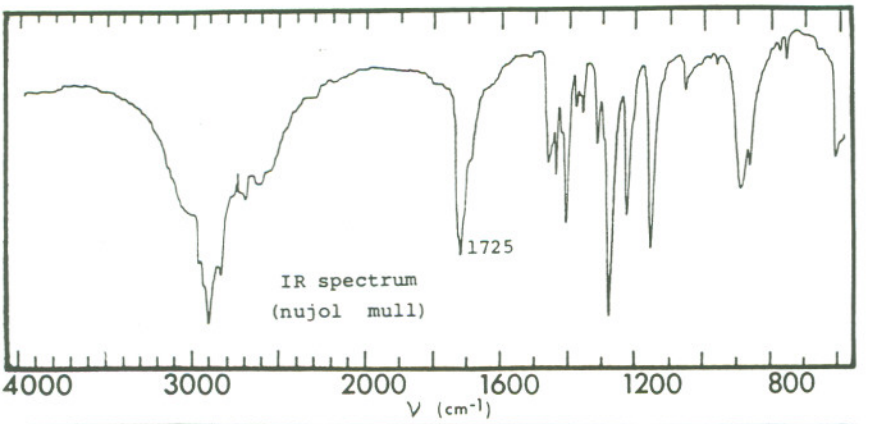

IR spectrum
(liquid film)
1720
4000
3000
2000
1600
1200
800
ν (cm⁻¹)

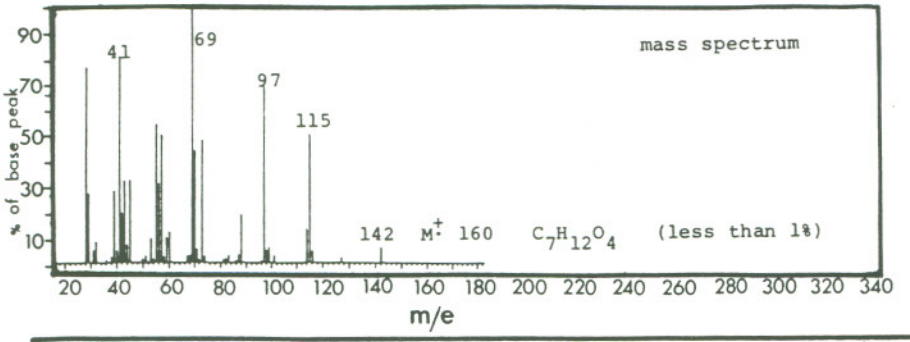

mass spectrum
105
91
77
M+· 212
% of base peak
90
70
50
30
10
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e

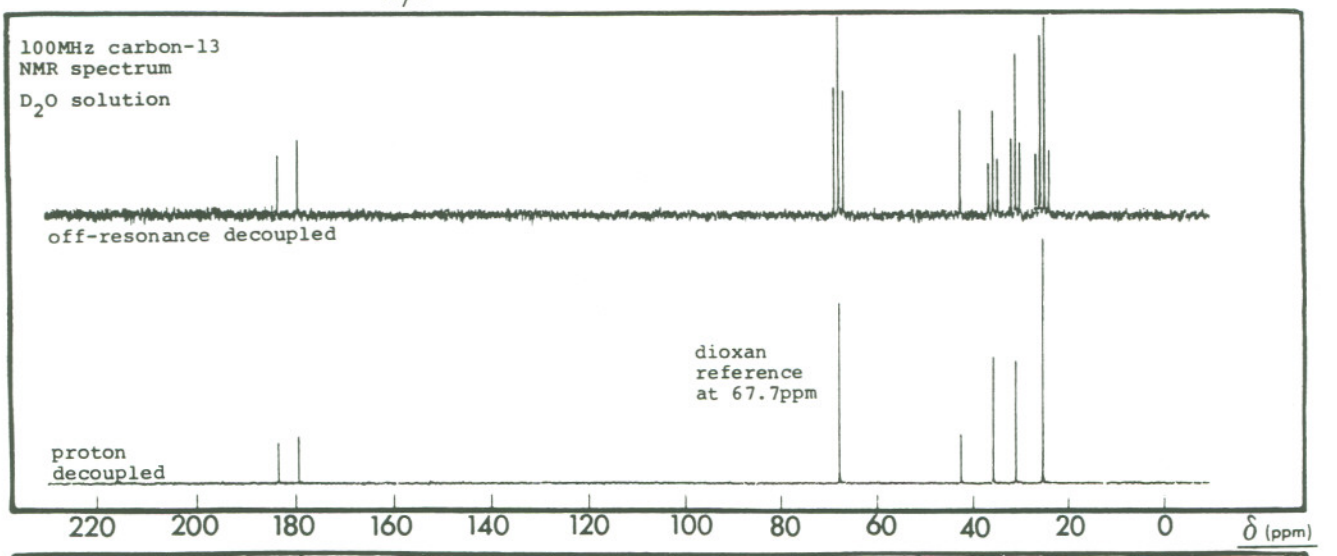

20MHz carbon-13
NMR spectrum
CDCl3 solution
resolves
into two
peaks at
higher
field
138
124
5x expansion
proton decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ (ppm)

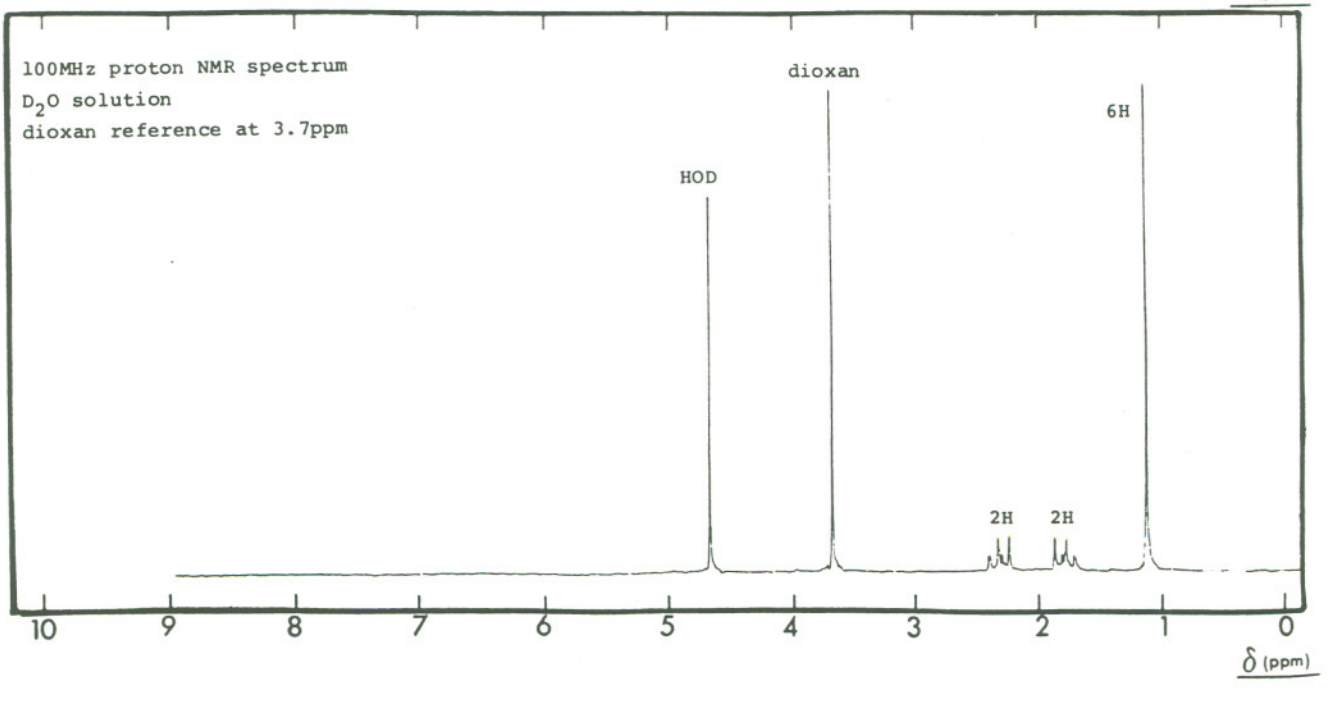

100MHz proton NMR spectrum
CDCl3 solution
10 9 8 7 6 5 4 3 2 1 0
δ (ppm)

Problem 3


1725
IR spectrum
(nujol mull)
4000
3000
2000
1600
1200
800
ν (cm⁻¹)


mass spectrum
% of base peak
90
70
50
30
10
41
69
97
115
142
M⁺ 160
C7H12O4
(less than 1%)
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e


100MHz carbon-13
NMR spectrum
D2O solution
off-resonance decoupled
dioxan
reference
at 67.7ppm
proton
decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ (ppm)


100MHz proton NMR spectrum
D2O solution
dioxan reference at 3.7ppm
dioxan
6H
HOD
2H
2H
10 9 8 7 6 5 4 3 2 1 0
δ (ppm)